

Understanding How Orthogonality of Parameters Improves Quantization of Neural Networks

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Abstract—We analyze why the orthogonality penalty improves quantization in deep neural networks. Using results from perturbation theory as well as through extensive experiments with Resnet50, Resnet101, and VGG19 models, we mathematically and experimentally show that improved quantization accuracy resulting from orthogonality constraint stems primarily from reduced condition numbers, which is the ratio of largest to smallest singular values of weight matrices, more so than reduced spectral norms, in contrast to the explanations in previous literature. We also show that the orthogonality penalty improves quantization even in the presence of a state-of-the-art quantized retraining method. Our results show that, when the orthogonality penalty is used with quantized retraining, ImageNet Top5 accuracy loss from 4- to 8-bit quantization is reduced by up to 7% for Resnet50, and up to 10% for Resnet101, compared to quantized retraining with no orthogonality penalty.

Index Terms—Deep neural networks, orthogonality regularization, perturbation theory, quantization.

I. INTRODUCTION

DEEP convolutional neural networks (CNNs) have achieved impressive results on a wide range of computer vision tasks from object classification, detection, segmentation, to image and video editing and interpolation. To achieve state-of-the-art results on these tasks, CNNs have become larger and deeper with increased complexity. On the one hand, the excellent accuracies coming from computationally heavy CNNs make them demanding to be used in various applications. On the other hand, it calls for computationally efficient implementations. There are several research directions to make CNNs run in a computationally efficient manner such as efficient implementations of CNNs on different hardware [1]–[6], novel convolutional neural network architectures that exploit memory and computation efficiency [7]–[11], knowledge distillation to decrease the number of parameters [12]–[15], pruning techniques to decrease the network size [16]–[18], and weight or activation quantization of CNNs from 32-bit floating point into lower bit-width representations [19]–[22].

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In this work, we are interested in network quantization to reduce the model size. Besides reducing the model size, low-precision quantization of weights and activations takes advantage of the low bit-width fixed-point arithmetic units on hardware [23], [24], resulting in power, area, and latency savings. These costs roughly go down linearly for addition operations with the number of bits and quadratically for multiplication operations. Model size, on the other hand, goes down linearly with the bit-width of quantized weights. Consequently, low-precision quantization is a valuable technique for enabling the highly efficient execution of CNNs. However, reduced precision commensurately reduces the number of representable values. This causes perturbations for each value of the input, weight, and output of every quantized layer at inference time, and these perturbations are carried on to subsequent layers, resulting in loss of accuracy, which can be significant for sufficiently low bit-widths (especially for less than 8 bits). Quantizing model weights and activations of a model trained in high precision (32-bit or 16-bit floating point) requires additional processing either during the initial training stage, a retraining stage, or post-training processing stage to mitigate the accuracy loss [24]–[26]. Recently, a retraining method for quantization was introduced [25], where a pretrained model is retrained with quantization thresholds included in the model parameters together with weights, where the quantization thresholds are trained simultaneously with gradient descent for optimizing the range of representable values after quantization. This method achieves state-of-the-art results for several networks and is called trained quantization thresholds (TQTs).

In a different research line, orthogonality in weights has proved to be beneficial for different purposes, such as making optimization more efficient [27], [28] or improving the generalization of the resulting models [28], and improving the quality of images synthesized by generative adversarial models [29]. Orthogonality constraint has also been used for improving the robustness of neural networks to adversarial examples [30], [31], with the latter work also studying robustness in the context of quantization, albeit with small datasets, such as CIFAR-10 or CIFAR-100. Alizadeh *et al.* [32] implements a second-order optimization method for imposing orthogonality constraints on weights, but this method increases training time by an order of magnitude, and it is only studied for small models such as Resnet18. Also, their work does not study whether their method is still beneficial when combined with trained quantization methods, or whether there are cases where trained quantization can

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